Yulia Mukhina

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Academic Positions

2026–now **Postdoc**, *University of Copenhagen*, Denmark
Applied Algebra and Geometry research group

2023–2026 **PhD student**, *École polytechnique*, France
Laboratoire d'informatique, MAX team. Advisors: Gleb Pogudin, Joris van der Hoeven

2022–2023 **Research assistant**, *Higher School of Economics*, Saint Petersburg
International Laboratory of Game Theory and Decision Making

Education

2023–2026 **Ph.D. in Computer Science**, *Institut Polytechnique de Paris*, École polytechnique
Advisors: Gleb Pogudin, Joris van der Hoeven
Thesis: "*Input-output relations: structure, computation, and applications*"

2017–2023 **Specialist (equivalent to Master's) in Fundamental Mathematics and Mechanics**, *Moscow State University*
Faculty of Mechanics and Mathematics. Advisors: E. I. Bunina, V. I. Zhukovskiy

Grants & Awards

2025 **SIAM AG25 Travel Award**, SIAM Conference on Applied Algebraic Geometry

2022 **BASIS Foundation Scholarship** for students of the Faculty of Mechanics and Mathematics, Moscow State University
Foundation for the Advancement of Theoretical Physics and Mathematics "BASIS"

Publications

Preprints

2026 **Engineered complete intersections: algorithmic aspects**
joint with A. Esterov, R. Mohr
Preprint · arXiv:2607.23622

Peer-reviewed articles

2026 **Projecting dynamical systems via a support bound**
joint with G. Pogudin
Mathematics of Computation · doi:10.1090/mcom/4248 · arXiv:2501.13680

2025 **Support bound for differential elimination in polynomial dynamical systems**
joint with G. Pogudin
Computer Algebra in Scientific Computing (CASC 2025) · doi:10.1007/978-3-032-09645-6_15 · arXiv:2506.08824

2025 **On the computation of Newton polytopes of eliminants**
joint with R. Mohr
Proceedings of the 2025 International Symposium on Symbolic and Algebraic Computation (ISSAC 2025) · doi:10.1145/3747199.3747564 · arXiv:2502.05015

2023 **Guaranteed solution for risk-neutral decision maker: an analog of maximin in single-criterion choice problem**
joint with V. Zhukovskiy, L. Zhukovskaya, S. Samsonov
Taurida Journal of Computer Science Theory and Mathematics, Vol. 2, pp. 7–30 · link

2023 **A new approach to guaranteed solutions of multicriteria choice problems: Pareto consideration of Savage–Niehans risk and outcomes**
joint with V. Zhukovskiy, L. Zhukovskaya
Taurida Journal of Computer Science Theory and Mathematics, Vol. 1, pp. 42–61 · doi:10.5281/zenodo.8102397

2023 **A new approach to optimal solutions of noncooperative games: accounting for Savage–Niehans risk**
joint with V. Zhukovskiy, L. Zhukovskaya
Taurida Journal of Computer Science Theory and Mathematics, Vol. 1, pp. 19–41 · doi:10.5281/zenodo.8102373

2021 **A differential game of N persons in which there is Pareto equilibrium of objections and counterobjections but no Nash equilibrium** (in Russian)
joint with V. Zhukovskiy, V. Romanova
Izvestiya Instituta Matematiki i Informatiki Udmurtskogo Gosudarstvennogo Universiteta, Vol. 57, pp. 104–127 · doi:10.35634/2226-3594-2021-57-04

2020 **Linear quadratic game of N persons as the analog of antagonistic game** (in Russian)
joint with V. Zhukovskiy, S. Samsonov, V. Romanova, L. Zhukovskaya
Taurida Journal of Computer Science Theory and Mathematics, Vol. 4, pp. 56–82 · link

Software

DiffMinPoly.jl – Julia package for the differential elimination problem for polynomial dynamical systems, · code on GitHub

Talks

Invited talks

Oct 2026 **Differential elimination in polynomial dynamical systems via a support bound**
Les Journées 2026 de l'Axe EFI, Institut Fourier, Grenoble, France

Sep 2026 **New elimination algorithms for differential dynamical systems**
Nonlinear Algebra Seminar, Max Planck Institute, Leipzig, Germany

Jul 2026 **Differential elimination: bounds, algorithms, and structure**
Foundations of Computational Mathematics (FoCM 2026), University of Vienna, Austria

May 2026 **Differential elimination: bounds, algorithms, and structure**
Differential Algebra and Related Topics (DART 2026), Île d'Oléron, France

Jan 2026 **Elimination for differential dynamical systems**
PolSys Seminar, LIP6, Sorbonne Université, Paris, France

Dec 2025 **Elimination algorithms for differential dynamical systems**
Technische Universität Braunschweig, Germany

Nov 2025 **Elimination for differential dynamical systems**
ODELIX – Thematic fall program 2025, LIX, École polytechnique, France

Oct 2025 **Differential elimination meets polyhedra**
Workshop "Algebraic methods in dynamics and particle physics", MPI for Software Systems, Saarbrücken, Germany

Jul 2025 **New algorithm for differential elimination based on support bound**
Applications of Computer Algebra (ACA 2025), Heraklion, Greece

Jul 2025 **Mixed fiber polytopes** (joint lecture with Rafael Mohr)
SLMath Summer School "New perspectives on discriminants and their applications", MPI, Leipzig, Germany

Jun 2025 **Differential elimination for dynamical systems**
Université de Limoges, France

Apr 2025 **A new algorithm for differential elimination**
Algebra and Representation Theory Seminar, Universität Bonn, Germany

Mar 2025 **Elimination in polynomial dynamical systems via a support bound**
Kolchin Seminar in Differential Algebra

Mar 2025 **Elimination for differential dynamical systems**
MAX computer algebra seminar, LIX, École polytechnique, France

Feb 2025 **On differential elimination algorithms**
Theoretical Cosynus Seminar, LIX, École polytechnique, France

Contributed talks

Mar 2026 **Forcing block-matrix structure for differential elimination**
Journées Nationales de Calcul Formel (JNCF 2026), CIRM, Marseille, France

Nov 2025 **Support bound for differential elimination in polynomial dynamical systems**
Computer Algebra in Scientific Computing (CASC 2025), Plekhanov University, Dubai, UAE

Jul 2025 **On the computation of Newton polytopes of eliminants**
ISSAC 2025, CIMAT, Guanajuato, Mexico

Mar 2025 **Elimination in polynomial dynamical systems via a support bound**
Journées Nationales de Calcul Formel (JNCF 2025), CIRM, Marseille, France

Jun 2024 **Support of the minimal polynomial in the differential dynamical system**
Encuentro de Álgebra Computacional y Aplicaciones (EACA 2024), Madrid, Spain

Apr 2023 **Niehans–Savage minimax regret principle in linear-quadratic problems**
Conference "Lomonosov", Moscow State University, Russia

Oct 2022 **On vector guarantees in multiobjective problems under uncertainty**
Tikhonov Readings, Moscow State University, Russia

Sep 2022 **Savage–Niehans principle for linear-quadratic problem**
Crimean Autumn Mathematical School-Symposium

Jun 2022 **Decision making based on the Pareto union of the principles of minimax regret and guaranteed outcome**
Conference "Differential Equations and Optimal Control", Steklov Mathematical Institute, Moscow, Russia

Apr 2022 **Savage principle and consideration of the outcome in a single-criterion nonlinear problem with uncertainty**
Lomonosov Readings, Moscow State University, Russia

Oct 2021 **A multiplayer differential game in which there is Pareto equilibrium of objections and counterobjections and no Nash equilibrium**
Conference for Young Scientists in Mathematical Economics and Economic Theory (MEET-2021), HSE, Saint Petersburg, Russia

Posters

Apr 2026 **Bounding the support in the differential elimination problem**
Future Algorithms 2026

Feb 2025 **Bounding the support in the differential elimination problem**
Special week "Enumerative combinatorics and effective aspects of differential equations", CIRM, Marseille, France

Jul 2024 **Bounding the support in the differential elimination problem**
MEGA 2024, MPI, Leipzig, Germany

Teaching

University

2025–2026 **Teaching assistant**, *École polytechnique* (lab sessions)
CSC-2S004-EP Machine learning; CSC-41011-EP Les bases de la programmation et de l'algorithmique (in French); CSE101 Computer Programming

2024–2025 **Teaching assistant**, *École polytechnique* (lab sessions)
CSE101 Computer Programming (Introduction to Python)

2023–2024 **Teaching assistant**, *École polytechnique* (lab sessions)
CSE102 Computer Programming (Introduction to Python)

Schools and mathematical circles

2020–2023 **Teacher of additional mathematics education**, *School 1514*, Moscow

2017–2020 **Tutor**, *Mathematical Circle of the Faculty of Mechanics and Mathematics*, Moscow State University

Supervision

2026 Jingyan Zhang (École Polytechnique), Bachelor thesis "*Reconstructing latent variables using integrals for epidemiological models*", co-supervised with G. Pogudin

2025 Maxime Despréaux (École Polytechnique), Bachelor thesis "*Faster linear algebra for differential elimination*", co-supervised with G. Pogudin

Professional Service

Reviewer ISSAC 2025, ISSAC 2026; *Journal of Symbolic Computation*

Skills

Programming Python, Julia, C/C++, Java, SQL

Software Maple, Macaulay2, SageMath, LaTeX, Linux

Languages Russian (native), English (fluent), French (fluent), German (beginner)